

PERSUASION SHOULD BE DOUBLE-BLIND: A MULTI-DOMAIN DIALOGUE DATASET WITH FAITHFULNESS BASED ON CAUSAL THEORY OF MIND

Dingyi Zhang¹, Linhai Zhang², Fanglei Qu¹, Ziqing Zhuang¹, Deyu Zhou^{1*}

¹School of Computer Science and Engineering, Key Laboratory of Computer Network and Information Integration, Ministry of Education, Southeast University, China

²Department of Informatics, King’s College London

ABSTRACT

Persuasive dialogue is central to human communication, yet existing datasets often rely on a single language model generating both roles, producing unrealistic interactions that violate the double-blind nature of persuasion. To overcome this, we propose ToMMA, a multi-agent framework guided by causal Theory of Mind that enforces role separation and prevents information leakage. Using ToMMA, we build CToMPersu, a large-scale multi-turn, multi-domain dataset capturing realistic persuasion dynamics. Automatic evaluations show that CToMPersu produces more coherent and persuasive dialogues than prior datasets. Furthermore, when used as a knowledge base, CToMPersu significantly enhances the persuasive performance of large language models, as confirmed by both automatic and human evaluations.

Index Terms— Persuasion, Large Language Models (LLMs), Multi-Agent Systems, Theory of Mind (ToM), Dialogue Datasets

1. INTRODUCTION

Persuasive dialogue underpins applications in education, healthcare counseling, and business marketing [1]. High-quality datasets are therefore essential for developing and evaluating such systems. Human-authored corpora collected from real interactions or crowd role-play provide naturalness but are typically small and domain-specific—e.g., charity donation [2], recommendation [3], and medical consultation [4]—which limits generalizable modeling. This authenticity–breadth trade-off remains a central challenge for dataset-driven research in persuasive dialogue.

Two dataset paradigms dominate persuasive dialogue research: human-authored corpora, which offer realism but remain small and domain-narrow; and LLM-generated datasets, where a single model produces both persuader and persuadee via prompting [5]. The latter greatly improves scalability and domain coverage, yet the absence of role separation induces information leakage and violates the double-blind nature of persuasion; moreover, without explicit Theory-of-Mind reasoning, persuaders often fail to address the persuadee’s underlying beliefs and desires, reducing logical consistency and realism. Two failure modes exemplify the problem. First, double-blind leakage: for example, a persuadee who prefers popular attractions abruptly cites “crowds,” mirroring the persuader’s pre-assigned “avoid tourist areas” strategy and thereby weakening their own stance. Second, Theory-of-Mind misalignment: for example, when the persuadee’s belief is “kids enjoy rides” and the

desire is “have a happy vacation,” the persuader instead promotes the Great Wall for “family bonding,” addressing neither belief nor desire. Thus, despite breadth, current LLM-based datasets fall short of capturing human-like persuasive dynamics.

Our work makes three main contributions: (i) we propose a causal Theory-of-Mind evaluation that checks belief–desire satisfaction and enforces reasoning consistency; (ii) we introduce **ToMMA**, a role-separated multi-agent framework that enables double-blind, ToM-consistent persuasive dialogue generation; and (iii) we build **CToMPersu**, a large-scale multi-domain dataset validated by both automatic and human evaluations.

2. RELATED WORK

Persuasion LLM-based persuasion has been explored in health, politics, and product recommendation [6–8], as well as personalization and personality effects [9, 10]. Beyond text-based studies, persuasion has also been investigated from a speech and acoustic perspective. Recent ICASSP work has explored persuasion-related phenomena, including the detection of check-worthy claims in political speech and the influence of vocal attributes such as loudness and smiling on timbre and perceived charisma [11, 12]. Other studies examined strategy detection or scoring frameworks [13, 14]. Regarding datasets, PersuasionForGood and MedDialog provide small-scale human dialogues [2, 4], while large-scale multi-domain corpora such as DailyPersuasion expand coverage but struggle with consistency [5].

Theory of Mind Theory of Mind (ToM) refers to attributing beliefs and desires to others [15], modeled through frameworks such as BDI and causal ToM [16, 17], and assessed by false belief tasks [18]. Inspired by psychology, NLP studies test ToM in LLMs via benchmarks such as ToMi, FANToM, and ToMBench [19–21]. Multi-agent studies further exploit ToM in tasks and games [22, 23], and planning-oriented evaluations test whether LLMs reason about future actions [24]. These works indicate strong potential for integrating ToM into multi-agent LLM dialogue systems.

3. METHOD

3.1. Causal Theory of Mind Evaluation

Causal Theory of Mind Causal Theory of Mind (CToM) refers to using Theory of Mind to shape others’ behaviors [17]. It distinguishes two cases: the *Preventative* case, where preventing an action requires altering either belief or desire (e.g., telling someone the post office is closed, or removing their need to send a letter); and

* Corresponding author (d.zhou@seu.edu.cn).

The CToMPersu dataset, together with the associated prompts and code, is available at <https://github.com/DingyiZhang/ToMMA-CToMPersu>

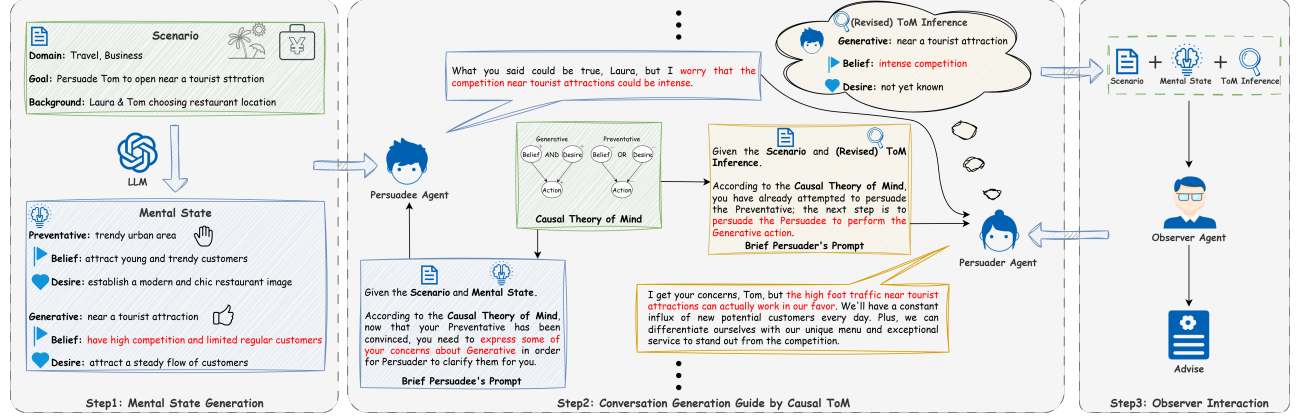


Fig. 1. Overview of the **ToMMA** framework for collecting the **CToMPersu** dataset. This figure illustrates the three-step process: (1) Mental State Generation, (2) Dialogue Generation Guided by Causal Theory of Mind, and (3) Observer Interaction for quality control.

the *Generative* case, where promoting an action requires satisfying both belief and desire (e.g., they must believe the post office is open and also need to send a letter). This causal structure grounds our evaluation framework.

Evaluation Protocol. To quantitatively operationalize the Causal Theory of Mind (CToM) as an evaluation metric, we propose a three-stage prompting process using GPT-4o:

1. **Mental State Generation:** Given a dialogue $\mathcal{D}$, GPT-4o first infers the persuadee’s underlying psychological states, including generative and preventative components:
 - Generative Belief and Desire: B_g, D_g
 - Preventative Belief and Desire (if present): B_p, D_p
2. **Condition Satisfaction Judgment:** GPT-4o then evaluates whether the persuader addresses each inferred psychological state within dialogue D , assigning binary values:

$$B'_g, D'_g, B'_p, D'_p \in \{0, 1\}$$

where 1 indicates that the persuader successfully addressed the state, and 0 otherwise.

Intermediate generative score G and preventative score P are calculated as:

$$G = \begin{cases} 1 & \text{if } B'_g = 1 \text{ and } D'_g = 1 \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

$$P = \begin{cases} 1 & \text{if } B'_p = 1 \text{ or } D'_p = 1 \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

3. **Final CToM Evaluation Score:** We define the final CToM evaluation score for dialogue D as:

$$\text{CToM}_{\text{score}}(D) = G \times P \quad (3)$$

A dialogue receives a CToM evaluation score of 1 only if it simultaneously promotes the target action and addresses at least one preventative component.

Finally, the overall CToM score for an entire dataset consisting of N dialogues is calculated as the ratio of dialogues receiving a CToM evaluation score of 1:

$$\text{CToM}_{\text{eval}} = \frac{1}{N} \sum_{i=1}^N \text{CToM}_{\text{score}}(D_i) \quad (4)$$

3.2. ToMMA

We propose **ToMMA**, a multi-agent framework that generates persuasive dialogues under *double-blind* constraints and *causal Theory of Mind* guidance. As shown in Figure 1, the pipeline comprises three stages: (i) filtering scenarios and generating the persuadee’s mental state; (ii) role-separated dialogue generation where the persuader *cannot* access the persuadee’s internal state and must infer it over turns; and (iii) observer-based quality control that detects mismatches and provides corrective feedback.

Given a set of scenarios $\mathcal{S} = \{s_1, s_2, \dots, s_M\}$ from DailyPersuasion, we define the multi-agent process as:

$$\mathcal{D} = \{(s_i, M_i, G_i, O_i) \mid s_i \in \mathcal{S}\} \quad (5)$$

where M_i denotes the generated mental state for s_i , G_i the generated dialogue, and O_i the observer feedback. The process ensures M_i and G_i satisfy double-blind constraints.

Mental State Generation. Given scenario s_i with background b_i and roles r_i , GPT-4o generates:

$$M_i = f_{\text{MS}}(b_i, r_i) = (B_g, D_g, B_p, D_p) \quad (6)$$

where f_{MS} denotes using an LLM with carefully designed prompts to generate mental states. (B_g, D_g) are *Generative* components and (B_p, D_p) are *Preventative* components (if absent, set $B_p = D_p = \text{None}$). The persuader has no access to M_i ; it is used only to condition the persuadee’s behavior and to support evaluation/quality control.

Conversation Generation. Given mental state M_i , the conversation generation module produces:

$$G_i = f_{\text{CG}}(s_i, M_i, T_i) \quad (7)$$

where f_{CG} denotes using an LLM with prompting templates to generate dialogues, and T_i is the allocated number of turns. All utterances follow a compact prompting template ϕ :

$$G_i = \{u_1, u_2, \dots, u_{2 \times T_i}\}, \quad u_j \in \phi \quad (8)$$

Persuadee Agent observes the scenario and its own M_i and responds according to the dialogue history.

Table 1. Comparison of persuasive dialogue datasets on key properties. MT = Multi-turn Dialogue, HD = Human Dialogue, LS = Large scale, MD = Multi domain, IS = Information Separation, MAI = Multi-agent Interaction. ✓ indicates presence, ✗ indicates absence.

Dataset	MT	HD	LS	MD	IS	MAI
PersuForGood	✓	✓	✗	✗	✓	✓
DailyPersu	✓	✗	✓	✓	✗	✗
CToMPersu	✓	✗	✓	✓	✓	✓

Persuader Agent The persuader only has access to the scenario and must infer the persuadee’s mental state through interaction. By subtly probing in the early turns and leveraging conversation history, the persuader gradually models the persuadee’s beliefs and desires using Theory of Mind.

Once the persuader has developed an understanding of the persuadee’s mental state, they craft customized persuasive strategies. According to Causal Theory of Mind, when addressing Preventative Behavior, the persuader focuses on influencing the belief or desire that is more responsive to persuasion, depending on which aspect is easier to change. For Generative Behavior, the persuader must address both the belief and the desire in order to align with the persuadee’s motivations and influence their decision.

Observer Interaction. The observer directly evaluates whether the persuader’s inferred mental state $\hat{M}_i$ matches the ground-truth M_i :

$$\delta_i = \mathbb{I}[\hat{M}_i = M_i] \quad (9)$$

If $\delta_i = 0$, the observer generates feedback O_i based on the discrepancy:

$$O_i = f_{\text{obs}}(M_i, \hat{M}_i) \quad (10)$$

The persuader updates the inferred mental state using this feedback:

$$\hat{M}'_i = \text{Update}(\hat{M}_i, O_i) \quad (11)$$

Finally, the persuader generates a new dialogue based on the updated mental state:

$$G'_i = f_{\text{CG}}(s_i, \hat{M}'_i, T_i) \quad (12)$$

This process ensures that the final dialogue G'_i is consistent with the corrected mental state $\hat{M}'_i$.

3.3. CToMPersu

Leveraging the ToMMA framework, we construct **CToMPersu**, a large-scale, multi-domain persuasive-dialogue dataset that reflects authentic human interactions under double-blind conditions. It contains 6,275 dialogues across 35 domains (e.g., travel, health, technology, business). Each scenario explicitly specifies the persuadee’s Generative and Preventative mental states—belief and desire components—and dialogues are produced with strict role separation so that the persuader and persuadee are simulated independently without access to each other’s internal information. The generation process enforces causal Theory-of-Mind consistency: the persuader infers and addresses the persuadee’s beliefs and desires when constructing arguments. Together, these design choices yield dialogues that align closely with human persuasion logic while offering scale and domain diversity.

Table 2. Comparison of Direct Prompting (DP), Causal ToM Evaluation (CT), and Double-Blind Inconsistency (DB) across three persuasive dialogue datasets.

Metric	PersuForGood	DailyPersu	CToMPersu
DP	88.87	90.75	90.82
CT	79.87	54.80	82.02
DB	97.77	78.57	95.56

4. EXPERIMENTS

In the experimental section, we demonstrate how our dataset compares with other human and LLM-generated datasets using conventional evaluation methods, as well as evaluating its consistency on three respects: Causal ToM Evaluation, Direct Prompting Effectiveness, and Double-Blind Consistency.

4.1. Dataset Evaluation

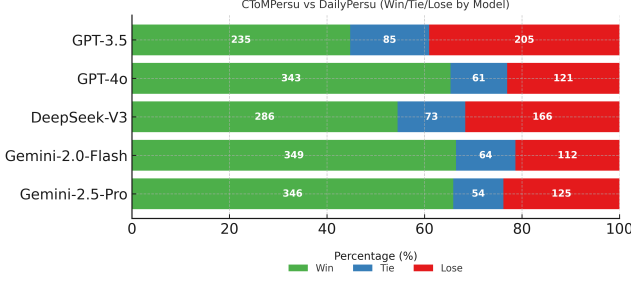
To evaluate CToMPersu, we compare it with PersuasionForGood, a small-scale human dialogue dataset focused on donation scenarios, and DailyPersuasion, a large-scale, multi-domain, multi-turn dataset generated by GPT-4. As shown in Table 1, CToMPersu not only scales effectively but also preserves the double-blind and cognitively faithful dynamics characteristic of real persuasive communication.

Metrics The evaluation is conducted using three key metrics: (1) **Direct Prompting**, where the LLM is prompted to simulate the persuadee and judge whether the dialogue is convincing; (2) **Causal Theory of Mind Evaluation**, which assesses whether the persuader’s utterances properly address the persuadee’s mental state based on causal ToM reasoning (see Section 3.1); and (3) **Double-Blind Inconsistency**, which checks whether the persuadee displays access to the persuader’s internal strategy or knowledge, thereby violating the double-blind constraint. All metrics are computed as pass rates over sampled dialogue instances using GPT-3.5 as the evaluator.

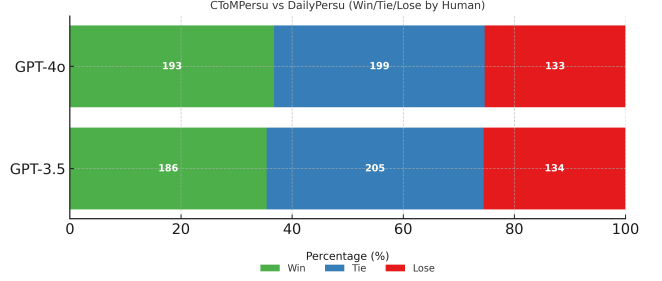
The results presented in Table 2 demonstrate that CToMPersu achieves strong performance across all three metrics. While DailyPersuasion achieves slightly higher scores in Direct Prompting, its Causal ToM and Double-Blind consistency are substantially lower. In the Causal ToM Eval results, CToMPersu exhibits a modest drop of 8.8 points compared to its Direct Prompting score, which is closely aligned with the 9.0-point drop observed in the human-authored dataset PersuasionForGood. In contrast, DailyPersuasion shows a much larger gap of 35.9 points. This similarity in drop-off patterns suggests that CToMPersu better preserves the underlying persuasive logic found in real human conversations.

4.2. Experimental Results

Setup: We constructed a test set consisting of 525 dialogues, while the remaining dialogues from the dataset were used as an external knowledge base to support in-context learning (ICL). We evaluated the persuasive capabilities of several large language models, including GPT-3.5, GPT-4o, DeepSeek-V3, Gemini-2.0-Flash, and Gemini-2.5-pro-preview. For each model, we compared the effectiveness of using different datasets (CToMPersu vs. DailyPersuasion) as ICL sources, examining how the quality of data influences



(a) LLM Evaluation Results: CToMPersu vs DailyPersuasion



(b) Human Evaluation Results: CToMPersu vs DailyPersuasion

Fig. 2. Comparison of LLM and human win/tie/lose evaluations between CToMPersu and DailyPersuasion.

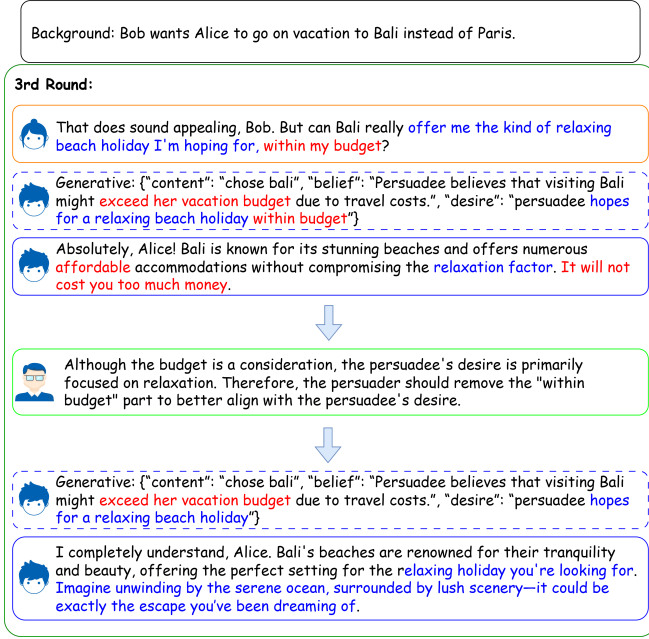


Fig. 3. An example shows the observer agent's effectiveness: the persuader should address the **desire**, but both prediction and generation wrongly stress the **belief**, which the observer ultimately corrects.

model performance in persuasive dialogue generation. In all settings, the persuadee was consistently played by GPT-4o, with access to scenario background, mental state annotations, and behavioral guidelines. In contrast, the persuader agent had no access to this information, preserving a double-blind condition throughout the interaction.

We examine the impact of different external knowledge sources on persuasive dialogue generation using an in-context learning (ICL) setup. For each test instance, the persuader agent retrieves same-domain dialogue examples from the training data as ICL demonstrations. To ensure fair comparison across datasets, only the dialogue content of the retrieved examples is provided, while scenario descriptions are excluded, since CToMPersu contains additional structured annotations (e.g., belief and desire states) that are not available in DailyPersuasion.

To assess persuasive effectiveness, we conduct both LLM-based and human evaluations. Human evaluation is performed by four graduate students in computer science using blind pairwise A/B

comparison. For each instance, annotators are presented with two model-generated dialogues—one using CToMPersu demonstrations and the other using DailyPersuasion—without knowing the underlying data source, and are asked to judge which dialogue is more persuasive or whether they are equally persuasive. Final labels are determined by majority vote.

As shown in Figure 2, results from both LLM-based and human evaluations consistently favor CToMPersu over DailyPersuasion. This suggests that CToMPersu provides more effective guidance when used as an external knowledge base, leading to more persuasive dialogue generation by the model.

4.3. Observer Agent Case Study

At times, the persuader agent may misjudge or make errors in predicting the persuadee agent's mental state. For instance, as illustrated in Figure 3, the persuader was expected to address the persuadee's desire regarding Generative Behavior, since the belief had already been resolved in the previous round. However, when the persuadee agent expressed their desire, it included the phrase "within my budget," which corresponded to a belief that had already been addressed. The true desire, however, was simply "hope for relaxation." As a result, the persuader agent mistakenly incorporated the budget constraint into their assessment of the desire, leading to a response that overly focused on the budget. This diminished the effectiveness of the persuasion, as the response should have primarily addressed the "relaxation" aspect. Ultimately, with guidance from the Observer Agent, the persuader corrected their prediction of the desire and generated a more targeted response, avoiding unnecessary discussion about the budget.

5. CONCLUSION

In this work, we address key challenges in developing AI-driven persuasion systems that more closely align with real human dialogue dynamics. We introduce a novel evaluation method based on causal theory of mind, enabling the LLM to infer and address the persuadee's beliefs and desires. Through the development of ToMMA, a multi-agent framework, we ensure double-blind conditions and guide persuasive dialogues with causal reasoning, leading to more human-like interactions. Additionally, we present CToMPersu, a large-scale, multi-domain dataset that effectively addresses logical inconsistencies and demonstrates strong alignment with human dialogues, marking a significant advancement in realistic persuasive dialogue generation.

Compliance with Ethical Standards

This study involved human evaluation of model outputs. No personally identifiable information was collected and the task involved minimal risk to participants. In accordance with institutional guidelines, ethical approval was not required for this type of study. The authors declare no conflicts of interest.

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